ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 120 cm	1.2 x 100 = 120.
2	**(c)** a mass vibrating on a spring	It moves to and fro on a straight path.
3	**(d)** hold the tape against the edge of the object so that measuring starts from zero	Hook sets zero.
4	**(b)** 3.5 km	Divide by 1000.
5	**(d)** circular motion	Path is circular.
6	**(b)** easier to read correctly	Accuracy.
7	**(a)** 50 cm	250 / 5 = 50.
8	**(d)** rotational motion	Rotates about its axis.
9	**(d)** 0.24 m	Divide by 100.
10	**(a)** 20 cm	80 / 4 = 20.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3.75 x 1000 = 3750 m (1); 3750 x 100 = 375000 cm (1).	1+1
12	Rotation: spinning about own axis, like Earth in a day (1); Revolution: going around another body, like Earth around the Sun (1).	1+1
13	Time period = total time / oscillations (1); 60/30 = 2 s (1).	1+1
14	4.5 m = 450 cm (1); 450 / 25 = 18 pieces (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	240 cm / 15 = 16 (1); 16 pieces exactly (1); Leftover 0 cm (1).	1+1+1
16	T = 42/21 = 2 s (1); 60 s / 2 s (1); 30 oscillations (1).	1+1+1
17	15 x 1000 = 15000 m (1); 15000 x 100 (1); 1500000 cm (1).	1+1+1
	OR: 1.1 m = 110 cm (1); 110 / 5 (1); 22 cm (1).	1+1+1
18	Wrap round the waist (1); Keep tape flat (1); Read where ends meet (1).	1+1+1
	OR: Pen: cm (1); River: km (1); Coin: mm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Wrap thread around the glass (1); Measure its length (1); Measure the diameter with a scale (1); Divide circumference by diameter (1); Ratio is nearly 3.14 (1).	1+1+1+1+1
	OR: Passenger at rest wrt train (1); Trees seem to move backward (1); Sun seems to move across the sky (1); Earth's rotation causes it (1); Depends on frame (1).	1+1+1+1+1
20	Sum = 601.5 (1); Mean = 601.5/5 = 120.3 cm (1); Repeating reduces errors (1); Readings vary slightly (1); Average is more reliable (1).	1+1+1+1+1
	OR: Hang two pendulums of different lengths (1); Count 10 oscillations (1); Record time (1); Longer takes more time (1); Conclude (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up conversions: 1 m = 100 cm (not 10 cm), 1 cm = 10 mm, 1 km = 1000 m.
- Reading the scale from the side, which gives a wrong value; keep the eye straight above the mark.
- Starting the measurement from the 1 cm mark without subtracting; either start at zero or subtract the first reading.
- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.

- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** orbital (revolution)	Revolution.
2	**(d)** 5	150/30.
3	**(a)** 0.1 mm	20 mm / 200 = 0.1 mm.
4	**(d)** 2 s	20 / 10.
5	**(d)** 2000 m	2 x 1000.
6	**(b)** 1.2 m	1.2 m = 120 cm is the greatest.
7	**(d)** the bus is moving relative to the tree	Relative motion.
8	**(d)** 800 m	Twice.
9	**(b)** 5 mm	5 mm.
10	**(a)** the lift	Same motion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Place a thread along the road (1); Measure thread and use the map scale (1).	1+1
12	150 cm (1); 1.5 m (1).	1+1
13	Zero mark may be missing (1); Can give wrong length (1).	1+1
14	Hold marble between two blocks (1); Measure distance between blocks (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	80 cm (1); 0.0009 km = 90 cm (1); 1 m = 100 cm, 1100 mm = 110 cm. Order: 80 cm, 0.0009 km, 1 m, 1100 mm (1).	1+1+1
16	Wrap thread around the edge (1); Mark the overlap (1); Straighten and measure (1).	1+1+1
17	Thread on line (1); Mark ends (1); Straighten and measure (1).	1+1+1
	OR: Wind the wire closely 20 times (1); Measure length of the coil (1); Divide by 20 for thickness (1).	1+1+1
18	5 alone has no meaning (1); 5 m differs from 5 cm (1); Units show the quantity (1).	1+1+1
	OR: Standard: fixed, same for all (1); Non-standard: vary, e.g. handspan (1); Results differ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Running: linear (1); Swing: periodic (1); Merry-go-round: rotational (1); Ball thrown: curved (1); Birds: random (1).	1+1+1+1+1
	OR: Straight: ruler (1); Curved: thread and ruler (1); Round objects: tape (1); Large distances: measuring wheel (1); Choose according to shape (1).	1+1+1+1+1
20	2.5 cm = 25 mm (1); 25 / 250 (1); = 0.1 mm (1); Because many sheets are measured (1); Gives more accuracy (1).	1+1+1+1+1
	OR: Measure the track length (1); Time the car (1); Longer distance in same time means faster (1); Repeat (1); Compare cars (1).	1+1+1+1+1

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- Stretching or loosening the thread while measuring a curved line.
- Calling a spinning top **circular** motion; spinning about its own axis is rotational.
- Saying all repeating motion is periodic; it must repeat after **equal** intervals of time.

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THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** measuring tape	Flexible.
2	**(c)** 7.5 cm	9.5 - 2.0.
3	**(b)** a straight line	Vertical line.
4	**(a)** slightly wrong	Expansion alters readings.
5	**(b)** 5.1 cm	Average = 15.3 / 3.
6	**(a)** random motion	Unpredictable.
7	**(d)** mm	Small length.
8	**(b)** 3.14	220 / 70 is about 3.14.
9	**(a)** rotational and linear (as it moves)	Wheel spins while moving ahead.
10	**(c)** metres	Large length.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Same everywhere (1); Calculations are easy with decimals (1).	1+1
12	Parallax error (1); Keep eye straight above the mark (1).	1+1
13	At rest relative to train (1); Moving relative to the platform (1).	1+1
14	Distance / 2 (1); 100/2 = 50 turns (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Straight: a falling stone (1); Curved: a thrown ball (1); Circular: hands of a clock (1).	1+1+1
16	Seated in a bus (1); Rest wrt seat (1); Moving wrt road (1).	1+1+1
17	Parallax (1); Zero error (1); Worn scale (1).	1+1+1
	OR: Direction and speed vary (1); No fixed pattern (1); Cannot be predicted (1).	1+1+1
18	Heartbeat: periodic (1); Moon revolving: periodic (1); Falling leaf, bouncing ball: non-periodic (ball bounces lower) (1).	1+1+1
	OR: Moves ahead (1); Changes direction in a curve (1); Wheels rotate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Body parts vary (1); Trade needed uniform units (1); Metre was chosen (1); Multiples km and cm (1); Accepted worldwide (1).	1+1+1+1+1
	OR: Eye at slant gives wrong mark (1); Eye straight above is correct (1); Use a thin scale (1); Place the scale close to the object (1); Repeat the reading (1).	1+1+1+1+1
20	3.6 km = 3600 m (1); 3600 / 1.8 (1); = 2000 turns (1); Distance/circumference (1); Each turn covers 1.8 m (1).	1+1+1+1+1
	OR: Cycle: linear (1); Wheels: rotational (1); Pedals: circular (1); Handle: oscillatory (1); Combination of motions (1).	1+1+1+1+1

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